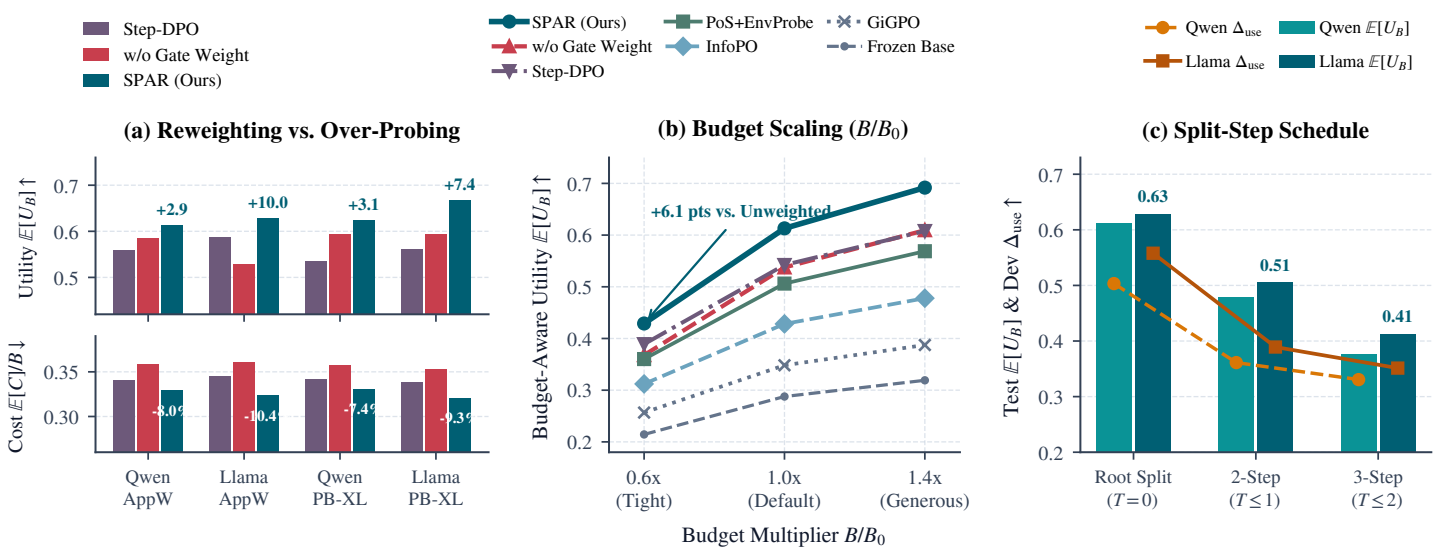
Step-DPO
w/o Gate Weight
SPAR (Ours)
SPAR (Ours)
w/o Gate Weight
Step-DPO
PoS+EnvProbe
InfoPO
GiGPO
Frozen Base
Qwen Δ_{use}
Llama Δ_{use}
Qwen $\mathbb{E}[U_B]$
Llama $\mathbb{E}[U_B]$
(a) Reweighting vs. Over-Probing
Utility $\mathbb{E}[U_B]\uparrow$
Cost $\mathbb{E}[C]/B\downarrow$
+2.9
+10.0
+3.1
+7.4
-8.0%
-10.4%
-7.4%
-9.3%
Qwen AppW
Llama AppW
Qwen PB-XL
Llama PB-XL
(b) Budget Scaling (B/B_0)
Budget-Aware Utility $\mathbb{E}[U_B]\uparrow$
+6.1 pts vs. Unweighted
0.6x (Tight)
1.0x (Default)
1.4x (Generous)
Budget Multiplier B/B_0
(c) Split-Step Schedule
Test $\mathbb{E}[U_B]$ & Dev $\Delta_{\text{use}}\uparrow$
0.63
0.51
0.41
Root Split ($T=0$)
2-Step ($T\leq 1$)
3-Step ($T\leq 2$)